CS 598 (Fall 2026)
Progress Report

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Introduction and Motivation:
Experts often have to make decisions based on noisy, partial summaries of cases, with budget-constrained visibility into the ground truth. In triage, for example, a doctor sees the patient's reported symptoms and vitals, which are noisy summaries of the patient's underlying ground truth state (complete biomarker measurements). Here, she must choose a set of tests to run, constrained by a limited budget, to uncover part of the patient's underlying condition or state before diagnosing and making a decision for what action to take. We study algorithms to help human decision-makers learn to make the correct decisions and become experts in such scenarios. We split this into two phases. During the training phase, an algorithm assists by selecting a set of features to show to the human, from which she learns to make the correct decision. Following this, in the graduation phase the human must select features on her own and must make a decision based on the revealed features. The algorithm seeks a curriculum that minimizes regret after graduation. We have derived optimal curricula for binary decisions with independent and correlated features, and extended the model to multiple actions. We then have worked on implementing these methods with real-world medical datasets, namely for sepsis prediction [5] and emergency triage diagnosis [3].

Related Work:
Our problem directly builds upon the framework introduced by Noti et al. [4] and extended by Guan et al. [1]. Noti et al. formulate the problem of an AI assisting a human decision-maker to learn the best coefficients for a linear prediction based on independent features, given a constrained number of features that can be shown. They find that stationary policies are optimal for human learning. Guan et al. extend this model, where the features are correlated and the human is able to impute values of unseen features. They find that policies which first explore and then commit to a fixed subset are optimal. We provide more related work and literature explored in the Appendix.

Unlike these papers, we model the human having to make a discrete decision based on features and observed symptoms, and look at the setting where the AI is not present after a fixed time. Thus the human must learn to make decisions on her own, unlike the previous works where the AI serves as a cognitive crutch—limiting human learning to the stationary subsets.

Feedback:
We received valuable feedback during both the in-class peer review and office hours. From the peer review session in class, we were advised to look into cognitive science works on how humans internally process information and try to map our "mental matrix" and learning curves to this. We also learned that we should make our motivation clearer and grounded in an example. Office hours were incredibly helpful in understanding what was previously done, avoiding solving problems that were already solved (namely bandits), and refining our problem formulation.

Work Done:
We formalize the problem of assisting human learning as follows:
Consider a human $\mathcal{H}$ who needs to take an action $k \in [K]$ based on an observation: $s \in \mathbb{R}^d$ of a hidden state $x \in \mathbb{R}^n$ drawn from distribution $\mathcal{D} = \mathcal{N}(0, \Sigma)$.

The ground truth utility matrix $U \in \mathbb{R}^{K \times n}$ with rows $u_1, ..., u_K$, gives us the utility of each move,

k, from the state x as $u_k^T x$. Thus the optimal action to make from x is $y^*(x) = \arg\max_k u_k^T x$. The human makes an action based on only an observation she sees, $s = Bx$, where $B \in \mathbb{R}^{d \times n}$ and a limited number of revealed true features $x_F := \{x_j : j \in F\}$ for $F \subseteq [n]$ with $|F| \leq c$. From the observation, she is able to estimate the true features: $x \mid s \sim \mathcal{N}(\mu(s), \Sigma_{x|s})$ where $\mu(s) = \Sigma B^\top (B\Sigma B^\top)^{-1} s$ and $\Sigma_{x|s} = \Sigma - \Sigma B^\top (B\Sigma B^\top)^{-1} B\Sigma$.

At time t, the human holds belief matrix $H^{(t)} \in \mathbb{R}^{K \times n}$. Thus from the observation $s^{(t)}$ and revealed features $x_F^{(t)}$, she holds the utility for action $j \in [K]$ as:

$$\hat{u}_k(F, x_F, s; H^{(t)}) = \sum_{j \in F} H_{kj}^{(t)} x_j + \sum_{j \notin F} H_{kj}^{(t)} \mu_j(s)$$

and takes action $\hat{y}\big(F, x_F, s; H^{(t)}\big) = \arg\max_{k \in [K]} \hat{u}_k(F, x_F, s; H^{(t)})$.

We look at the problem of an algorithm $\mathcal{A}$ assisting $\mathcal{H}$ in learning the true utilities, making the correct move, and selecting the correct set of features to reveal. This consists of two phases: a training phase and graduation phase.

The training phase occurs for $t = 1, ..., T$. Here:

- $x^{(t)} \sim \mathcal{N}(0, \Sigma)$ and $s^{(t)} = Bx^{(t)}$ are realized
- $\mathcal{A}$ sees $s^{(t)}$ and picks $F_t \subseteq [n]$ with $|F_t| \leq c$ to show to $\mathcal{H}$
- $\mathcal{H}$ observes $s^{(t)}$, and $(F_t, x_{F_t}^{(t)})$, makes move $\hat{y}\big(F_t, x_{F_t}^{(t)}, s^{(t)}; H^{(t)}\big)$ and observes the decision regret loss: $\ell_{\text{reg}}(x^{(t)}, \hat{y}_t) = \max_k u_k^T x^{(t)} - u_{\hat{y}_t}^T x^{(t)}$
- $\mathcal{H}$ updates her matrix, contracting towards the true utility for the features and actions observed: $H_{kj}^{(t+1)} - U_{kj} = \psi(m_{kj}(t))[H_{kj}^{(0)} - U_{kj}]$ for $(k, j) \in [K] \times F_t$, and where $m_{kj}(t) = |\{\tau < t : k \in [K], j \in F_\tau\}|$ and $\psi : \mathbb{N} \to [0, 1]$ is non-increasing with $\psi(0) = 1$. For unmeasured features, the beliefs remain the same.

The graduation phase occurs after time T, where for each $t > T$:

- $\mathcal{H}$ observes $s^{(t)}$ and picks subset $F_t = F_{\mathcal{H}}^*(s^{(t)}; H^{(T)})$ where F_t is chosen to maximize her perceived value of information:

$$F_{\mathcal{H}}^*(s; H^{(T)}) = \arg\max_{|F| \leq c} \mathbb{E}_{x_F|s} \left[\max_{k \in [K]} \hat{u}_k(F, x_F, s; H^{(T)}) \right]$$

- The true features $(F_t, x_{F_t}^{(t)})$ are revealed to $\mathcal{H}$.
- $\mathcal{H}$ makes move $\hat{y}_t = \hat{y}(F_t, x_{F_t}^{(t)}, s^{(t)}; H^{(T)})$ and observes loss: $\ell_{\text{reg}}(x^{(t)}, \hat{y}_t)$

The goal of the algorithm is to select training policy: $\pi = (F_1, F_2, ..., F_T)$ minimizing the combined objective:

$$\mathcal{L}(\pi) := \lambda \sum_{t=1}^{T} \mathbb{E}\big[\ell_{\text{reg}}(x^{(t)}, \hat{y}_t)\big] + \sum_{t>T} \delta^{(t-T)} \mathbb{E}\big[\ell_{\text{reg}}(x^{(t)}, \hat{y}_t)\big]$$

where $\lambda \in [0, 1)$ is small cost incurred ($\lambda \approx 0$) on the training phase, and $\delta \in (0, 1)$ is discount factor. We consider that $\mathcal{A}$ has full access to $U, H^{(0)}, \psi$.

So far, we have worked out the theory for the following:

- Binary decision case ($K = 2$):
  - Coordinate reduction in terms of differences and closed-form formula for $F^*_{\mathcal{H}}$
  - Expected regret formula and optimal single-feature ($c = 1$) solution for both independent and correlated features
  - Optimal offline curriculum allocation algorithm for any $c \geq 1$ for both independent and correlated features
- Multi-action case ($K \geq 2$):
  - Test selection and optimal schedule under multi-action correlated features
  - Extension where $\mathcal{A}$ can show/highlight $a_{\min} \leq A_t \leq a_{\max}$ actions as well as F_t, and derived optimal schedule
  - Formalizing cognitive forgetting via Restless Multi-Armed Bandit formulation, and optimal features $\mathcal{H}$ selects to minimize this (graduation time strategy)

Plan:
On the theory side, we are working on verifying the algorithms, theorems, and proofs for the multi-action settings (which are generated by Gemini)—to better understand and communicate them. At the same time, other members are working on real-world implementations of these algorithms and theories on selected medical datasets. We have already simulated a pipeline fitting our model to the open MIMIC-IV-ED demo [3] (166 ED stays, $K = 2$: admit vs. discharge, seven triage vitals as features, U estimated via logistic regression and Σ from empirical covariance). Our simulations reproduce the analytical graduation regret formula to within 1.2% across thirty belief states and panel sizes $c \leq 3$. Crucially, we observe that the human's autonomous test selection $F^*_{\mathcal{H}}$ switches to a near-zero-weight feature exactly when her misconception about it exceeds the c-th largest $|d_j|$, after which graduation regret exceeds that of a random choice. We aim to scale this to the full MIMIC-IV-ED (to use features of patient vitals as observed summary s, the hidden features, credentials are in progress) and an open-source dataset from PhysioNet 2019, Sepsis [5]. Based on these open data, we will 1) compare the offline curriculum allocation algorithm against exhaustive search and naive curricula; 2) test the ex-post policy when $\mathcal{A}$ observes the state; 3) examine the correlated feature case and forgetting case in the real world to explore how often they appear, how to simulate them, and how the algorithm can optimize against them. Following the peer review feedback, we also want to learn more about cognitive science theories on how humans update beliefs, and from which get a better understanding of $H^{(0)}$ and ψ, since no dataset currently supplies them. We are also curious about how sensitive the optimal curriculum is to this choice of modeling.

Questions: Is our formalized problem formulation clear and well-motivated? Are there any assumptions we have made which are too strong and might not hold for real settings? Are there any other interesting directions we should look into (on the theory side) to make this more realistic, given LLMs are able to work this theory out really well?

References

[1] Yanru Guan, Naveen Raman, and Fei Fang. Human decision-making with AI assistance under correlated features. *arXiv preprint arXiv:2606.20628*, 2026.

[2] Dylan Hadfield-Menell, Stuart J. Russell, Pieter Abbeel, and Anca Dragan. Cooperative inverse reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, 2016.

[3] Alistair E. W. Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J. Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, Li-Wei H. Lehman, Leo Anthony Celi, and Roger G. Mark. MIMIC-IV-ED, a freely accessible electronic emergency department database. *Scientific Data*, 10(1):1, 2023.

[4] Gali Noti, Kate Donahue, Jon Kleinberg, and Sigal Oren. AI-assisted decision making with human learning. In *Proceedings of the 26th ACM Conference on Economics and Computation (EC)*, 2025.

[5] Matthew A. Reyna, Christopher S. Josef, Russell Jeter, Surya P. Shashikumar, M. Brandon Westover, Shamim Nemati, Gari D. Clifford, and Ashish Sharma. Early prediction of sepsis from clinical data: The PhysioNet/computing in cardiology challenge 2019. *Critical Care Medicine*, 48(2):210–217, 2020.

[6] Xiaojin Zhu. Machine teaching: An inverse problem to machine learning and an approach toward optimal education. In *Proceedings of the 29th AAAI Conference on Artificial Intelligence (AAAI)*, pages 4083–4087, 2015.

Contribution Statement:
Olly: Conducted simulations on demo data, drafted future plan sections, and initiated dataset credentialing.
Chirag: Worked on developing problem model and algorithm implementation.
Steven: Worked on the related work section.
Eeshan: Worked on the theory and problem formulation.

LLMs were used throughout the process, in better understanding the problem, assisting in finding and solving various theorems and algorithms, as well as in literature search.

Appendix:
More Related Works:
We have been actively reviewing related papers throughout the project so far, and they shaped both how we formulated our research question and how we designed our methodology. Most works on algorithms that support human decision-makers treat the human as fixed: the algorithm recommends an action or supplies information, and the way that the human reads that information never changes. Our project falls into a new line of work in which we explore the case when the human learns from what the algorithm shows them.

Noti et al. [4], *"AI-Assisted Decision Making with Human Learning":* Noti et al. let an algorithm choose which features a human sees and show that, with independent features, showing the same

subset every round is optimal. We take our core model from them (feature budget, learning rule, exposure counts).

Guan et al. [1], *"Human Decision-Making with AI Assistance under Correlated Features":* Guan et al. show that with correlated features this fails and the problem becomes NP-hard. We take from them the human's estimate of unseen features (our $s = Bx$) and dynamic programming over exposure counts.

Zhu [6], *"Machine Teaching: An Inverse Problem to Machine Learning and an Approach Toward Optimal Education":* Zhu frames teaching as choosing the best examples for a known learner, which is how we view our algorithm. Unlike in machine teaching, our algorithm cannot choose the examples, and each one is also a real decision.

Hadfield-Menell et al. [2], *"Cooperative Inverse Reinforcement Learning":* In cooperative inverse reinforcement learning, a robot infers a human's goal, and acting worse now can teach more. Its learning-then-deployment structure parallels our training and graduation phases. Here the AI has to infer what the human knows, whereas in ours the AI already knows the ground truth and the human's beliefs are what need correcting.

Feature and Distribution Engineering:
Since real features are rarely normal, we create the above x and s as normal scores from each raw feature and raw observation. Consider raw feature r_j, with any CDF C_j, it becomes $x_j = \Phi^{-1}(C_j(r_j))$, where Φ is the standard normal CDF. Each raw observation is also converted the same way with its own CDF G_i. We assume only that the scores are jointly normal (a Gaussian copula with correlation matrix Σ), which loses only relationships that are not monotone.